

# Determining the relationship between the Q factor, period, and length of a pendulum

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## 1 Introduction

This lab explores the dynamics of a flashlight hung by a nylon string, modelled as a pendulum with damped harmonic motion. Particularly, the relationship between its period and initial angle of release, the period and the Q-factor, the period and the length, and the length and Q-factor, were explored and compared to the theory provided in (Wilson, 2024). The experimental data disagreed with all the theory provided. It was found that the relationship between period and initial angle of release was quadratic, with symmetry within the error bars. This disagrees with the theory provided in (Wilson, 2024), where the author states that the  $C$  value (and higher order values) in equation 1

$$T = T_0(1 + B\theta_0 + C\theta_0^2) \quad (1)$$

would be zero, where  $T$  is the period measured in seconds, and  $\theta$  is the initial angle of release, measured in radians. Experimentally, it was found that the  $C$  value was  $1.17 \pm 0.01$  making the quadratic term non-zero. The Q factor for the fixed length of  $36.5 \pm 0.2$  cm, which was determined using two methods. The first method was the counting method, from which the Q factor was determined to be  $96 \pm 4$ . The Q factor was also measured using a line of best fit in equation 2,

$$\theta(t) = \theta_0 e^{-t/\tau} \cos(2\pi \frac{t}{T} + \phi_0) \quad (2)$$

from which the Q factor was determined to be  $156 \pm 5$ . By comparing the error in the two values, it was determined that the line of best fit method was more consistent, so that was the value used. The period was found to have an exponential relationship

with the length of the pendulum, with an equation of

$$T = C \times L^x \quad (3)$$

where  $C = 2.03 \pm 0.04$ ,  $x = 0.36 \pm 0.03$ , and  $L$  is the distance from the pivot to the centre of mass of the pendulum, measured in metres. This is in disagreement with the theory provided in (Wilson, 2024), which stated that the relationship would have  $C = 2$  and  $x = 0.5$ . The Q factors were determined using the line of best fit method (Wilson, 2024) to determine  $\tau$ , and then equation 4 was used.

$$Q = \pi \times \frac{\tau}{T} \quad (4)$$

There were no relevant numerical results for the relationship between the length and Q factor. It is hypothesized that the data did not match the theory provided because of the presence of significant non-planar motion as well as energy losses, which are not reliably quantifiable at the level of the report.

## 2 METHODS

All experiments were performed with the same basic setup, as shown in figure 1. A nylon string mass  $< 1.0$  g, was attached to a flashlight mass  $= 121.0 \pm 0.1$  g and hung from a coat rack. The nylon string was chosen for its lightweight and relatively inelastic properties, two key experimental parameters discussed in class. Black electrical tape was placed on the attachment points of the string to minimize rotation and to restrict motion to the plane of pivot. Heavy weights (mass  $> 5$ kg) and electrical tape were placed to secure the coat rack, as initial

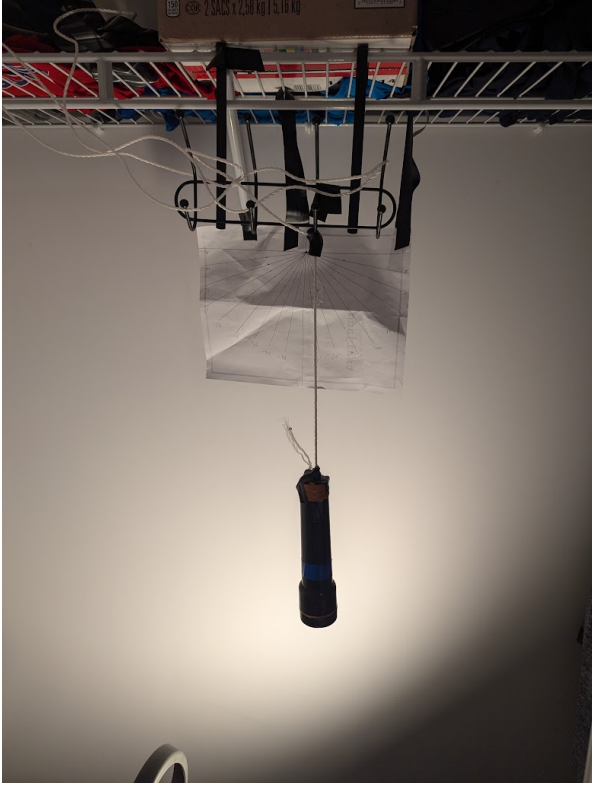


Figure 1: The experimental setup. Note the blue tape placed at the centre of mass and the heavy weights and tape placed on the coat rack to reduce unwanted motion.

pilot tests without any restrictions caused a lot of movement of the coat rack, indicating energy loss. A coat hanger was used to hold the flashlight before release, to ensure alignment with the initial angle and to ensure that the centre of mass of the flashlight was in the axis of the string.

The Autotracker function of the app *Tracker* was used to track the position of the centre of mass. It was suggested (Brown, 2024) that tracking is improved with a distinctly coloured feature on a solid-coloured background. Thus, a blue piece of electrical tape was placed on the centre of mass and the background lit up white to assist the tracking. The centroid of the flashlight was determined as the plane that I could balance on my finger, and the centre of mass was the middle of the centroid.

My laptop camera (720p, 30fps) was used to record each trial, with the centre of the image at the pivot point of the pendulum. Par-

allax errors were determined to be negligible. The periods were measured by observing each frame and finding when the pendulum reached its maximum displacement. This frame was determined when the displacement increased from the previous frame and decreases in the next frame, which was much easier to spot than at nadir, which was suggested (Wilson, 2024).

For lab 2, it was determined through the results of lab 1 that the period remained effectively constant at small angles ( $|\theta| < 40^\circ$ ). Therefore, to investigate the relationship between the period and the pendulum's length, the initial release angle was maintained below this threshold. Another result of lab 1 was that for the amplitude-time graph, as the amplitude got smaller, the exponential fit looked a lot like a linear fit. Thus, it was easier to find the Q factor with larger angles. These factors were used to determine the following experimental methods. As recommended in (Wilson, 2024), there were 8 measured lengths, spanning an order of magnitude at roughly equal intervals (from  $0.095 \pm 0.001\text{m}$  to  $0.775 \pm 0.001\text{m}$ ). To maintain the small angle approximation while also gathering useful data, for all trials the pendulum was released at  $+30^\circ$ , and measured until the amplitude was effectively 0. The period and Q factor were measured using the same methods as in lab 1.

### 3 RESULTS

The period-amplitude data (figure 2) was determined to have a quadratic fit. Using equation 1 from the introduction, the  $A$  value of the equation is  $0.11 \pm 0.01$ . Symmetry was tested by measuring the same positive and negative release angles, which gave results that were the same within the error bars. As well, the  $B$  value was determined to be  $0.01 \pm 0.01$ , which is experimentally zero, indicating that the pendulum is symmetric (Wilson, 2024).

The  $C$  value of the pendulum was  $1.17 \pm$

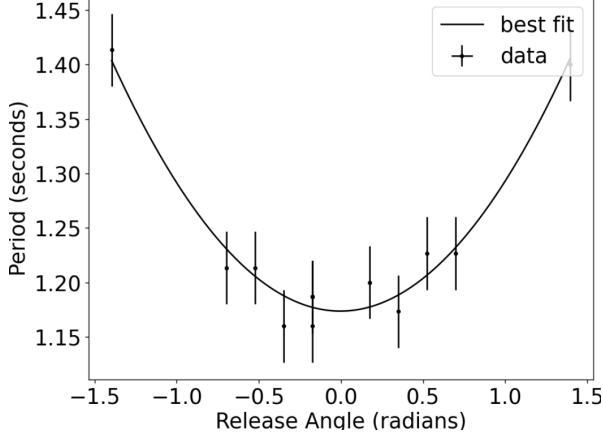


Figure 2: The period amplitude graph, which shows a quadratic fit, disagreeing with the theory of a linear relationship (Wilson, 2024). Symmetry can be seen in the data points, but also in the  $B$  term of equation 1. The horizontal error bars are too small to be seen, but have a value of 0.005 radians.

0.01, disagreeing with the theory that stated  $C = 0$  (Wilson, 2024). Thus, the relationship between the period and initial amplitude is quadratic, although at small angles  $|\theta| < 40^\circ$ , the period is experimentally constant within the error bars, as can be seen in figure 2.

The amplitude-time graph (figure 3) was determined to have an exponential fit. The Q factor was measured in two different ways, by counting the number of cycles until the amplitude diminished to 4% of its initial value and by the line of best fit method described in (Wilson, 2024). For the counting method, to minimize human error, a python code was implemented to determine the number of cycles it took for the amplitude to decay the desired value. When counted from the beginning, the Q factor was determined to be  $96 \pm 4\%$ , with the largest uncertainty being in the amplitude itself, giving an absolute uncertainty of  $96 \pm 4$ . The problem with the counting method was that the Q factor varied without pattern depending on where the initial amplitude was chosen. Thus, it was determined to be unreliable compared to the line of best

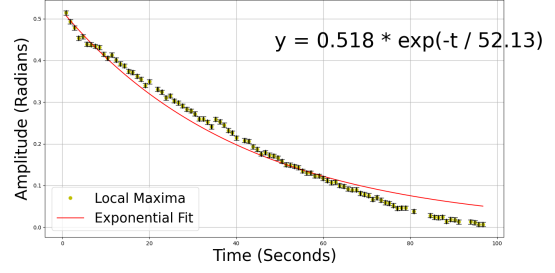


Figure 3: The amplitude-time graph with an exponential decay relationship. The value of  $\tau$  was determined to be  $52.13 \pm 4\%$ , giving a Q factor of  $156 \pm 5$ . By the counting method, the Q factor was determined to be  $96 \pm 4$ , although that value is a lot less consistent. The horizontal error bars are too small to be seen, but have a value of 0.03 seconds. Full data can be viewed in the code repository linked in the appendix.

fit method, which was more consistent as there was only one line of best fit. With this consideration, the line of best fit method was employed for lab 2. For the line of best fit method, rather than the suggested approach, the cosine term was ignored as it was irrelevant to the Q factor calculation. Instead, the maxima points were determined using a Savitsky-Golay filter (See Appendix A for code), and a line of best fit was passed through those points. This had the effect of reducing the noise, as the intermediate positions were not well tracked by *Tracker*, whereas the extrema position had a much lower uncertainty. As well, this effectively modified equation 2 to become equation 5.

$$\theta(t) = \theta_0 e^{-t/\tau} \quad (5)$$

The line of best fit method thus produces a Q factor of  $156 \pm 5$ .

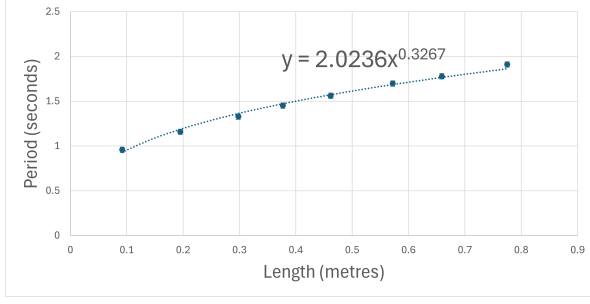


Figure 4: The period-length graph, with an exponential relationship. The horizontal uncertainties are too small to be seen, with a value of 0.001 metres. The vertical error bars are also essentially too small to be seen, and they have a value of 0.03 seconds. The value of the exponent is in disagreement with the theoretical value of 0.5.

To determine the relationship between the period and the length (figure 4), the same method was applied in determining the period as above. The length was found to have an uncertainty of  $\pm 0.001\text{m}$  for all results.

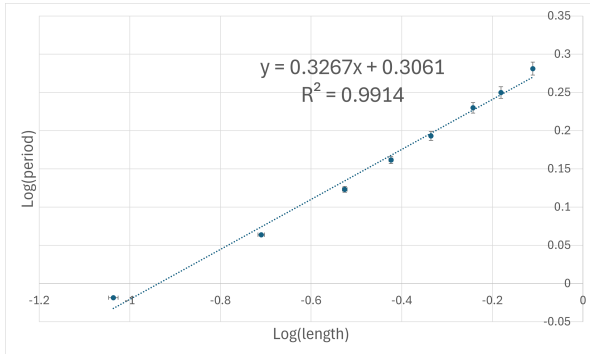


Figure 5: The log-log plot of the period of the pendulum (originally in seconds) and the length (originally in metres). The uncertainties were the original measured uncertainties, found in the exponential plot, converted to percent. It is easier to see that the best fit parameter of 0.39 is in disagreement with the theoretical value (Wilson, 2024), which is not within the error bars

The data was also plotted on a log-log graph (figure 5), where the best fit parameter can be easier to see. The percent uncertainty values remain constant, and the sources for uncertainty are the same as for the exponential plot, as it is the same experiment.

As previously mentioned, the line of best fit method was less prone to error, so it was used to calculate the Q factor for all the lengths. The lengths were the same as the ones used in the period-length plot, with the same measured uncertainty. The vertical error bars are the percent uncertainty in the period, as that was the largest source of uncertainty in the parameters of equation 4. As was determined by the period-length plot, the period varied for different lengths, so the percent uncertainty varied for each Q factor value. The actual Q factors themselves were calculated using the same method as the initial calculation in lab 1.

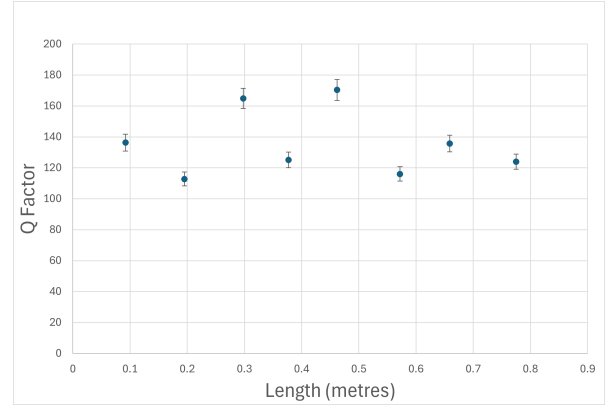


Figure 6: The Q factor-length graph. It can be clearly seen that no relevant relationship can be determined. The horizontal error bars are too small to be seen, but have a value of  $\pm 0.001\text{m}$

Evidently, there is no clear relationship between the Q factor and length (figure 6). On the surface, there appears that there could be a negative quadratic relationship, but the  $r^2$  value for such fits were well below 0.2, or statistically irrelevant. As there was no theoretical value to compare to, it was impossible to determine how much percent error there was in determining the relationship.

## 4 DISCUSSION

### 4.1 Uncertainties

The uncertainty in the period angle graph is type B uncertainty, where it is measured as the duration of one frame ( $\delta_{period} = 0.03$  seconds). The uncertainty is the same for all trials, as they were all recorded at 30 fps. This was different than the suggested value of  $0.25 \times \text{duration of frame}$ , which was discussed in class and in (Wilson, 2024). The suggested value was for the scenario that the exact frame of interest was known. This larger uncertainty value was chosen as there were values where the exact frame that contained the maxima was unclear (i.e. between two frames), which would quadruple the uncertainty of one frame, as in the worst case scenario, the value could be in any one of three frames.

The measured uncertainty in the angle was determined by using half the smallest increment of the protractor ( $\delta_{angle} = 0.009$  radians). The type A uncertainty was larger than this, so that was the value that was used. It was measured by repeating the trial at  $-0.175$  radians 5 times, and the largest difference in period was determined to be ( $\Delta_{period} = 0.4$ ). The standard deviation of the period calculated using this value was 0.01 which when plotted as horizontal error bars are too small to be seen. Once this uncertainty was calculated, it was decided that it was unnecessary to repeat trials for other values, as other uncertainties would be much larger and much more relevant.

There were two different uncertainties for the two different Q factor values. The line of best fit method produced an uncertainty that had a value of 3%, or for  $Q = 156$ ,  $\delta_Q = 5$ . The line of best fit method of determining Q factor was determined to be more accurate, as the values found counting method were dependent on the starting point of the count. These differences did not follow any recognizable pattern, it was impossible to quantify this error to anything physically meaningful. Sources for this un-

quantifiable error are discussed below. This meant that there were significant differences in the Q factor that was not reflected in the uncertainty, which was  $\delta_{Qfactorcounting} = 4$ . This uncertainty was not in the counting, but rather came directly from determining the value of amplitude/ $e^\pi$ , which was the same as the uncertainty in the amplitude, or 4%.

This also would have had the effect of reducing the Q factor. The Q factor is effectively a measure of the “quality” of a pendulum, and the more the system loses energy, the lower the Q factor. As it is outside the scope of this lab to quantify this loss of energy, there is a possibility that the effect of this energy loss is not consistent, which could explain the lack of relationship in Q factors between different lengths. As well, given the plastic nature of tape, it is possible that successive trials altered the set up, specifically the big friction producing parts such as the tape, making comparisons between trials off by an unknown factor. It is more likely that the former had a much bigger effect.

### 4.2 Errors and Improvements

It was observed that at especially higher amplitudes  $|\theta| > 40^\circ$ , the flashlight had a tendency to travel off axis, or “flutter” around in three dimensions. This would result in lower experimental amplitude measurements, as the non-harmonic motion requires energy, decaying energy quicker from the pendulum, meaning at the extrema, there is less potential energy and less height. During pilot trials without any motion restrictions, when the pendulum was released, the entire system shook. The attachment rack noticeably moved, and the flashlight was rotating, even with small release amplitudes. The solution to this was to place a heavy weight on top of the rack to minimize motion, and to put electrical tape on the moving parts that were intended to be stationary. The heavy weight worked well, as no movement of the attachment rack was visible.

The tape also worked in this regard, but had a side effect of introducing some friction to the system. The tape had plastic properties, so some energy was dissipated into the tape. As friction increases, the line of best fit of the amplitude-time graph becomes more linear (Wilson, 2024), which would reduce the calculated Q factor. Future improvements would include having a more rigid pivot attachment in order to restrict motion to the desired plane. While the methods described did visibly limit motion, there was still significant energy loss that was not accounted for in the damped harmonic model (Wilson, 2024). The model stated that adding friction would increase the linearity of the line of best fit, which did occur, and a suggested improvement in order to match the model better is to reduce friction. This source of error is infeasible to quantify at the level of this report; thus was not taken into consideration when deciding the size of error bars. *Tracker* works better with higher quality videos, as it tracks objects frame by frame (Brown, 2024). Motion in between frames causes image distortion, which the autotracker function has difficulties dealing with. The choice of a narrow object like a flashlight as the pendulum bob did help in this regard, as when it was blurred, the autotracker had fewer points to pick from, which meant that the point it did pick was close to the actual centre of mass. At lower amplitudes, the maximum speed is lower, so the frames are less blurry, resulting in better data. To improve tracking accuracy at higher pendulum speeds, it would be beneficial to use a higher resolution camera that recorded in more frames per second. A higher resolution camera would also allow for more detailed position markers on the pendulum, which would assist in tracking (Brown, 2024). The low-fidelity method used in this lab involved placing blue tape, which while did contrast with the black flashlight and white background, gave *Tracker* many blue pixels to choose a position from, lowering the overall precision of the position. Bypassing the lim-

itations of video would improve the tracking accuracy of the position; however this would mainly affect the intermediate positions between the positions of maximum displacement, and it will not significantly reduce the uncertainty in the position at the extrema, since those were relatively easier for *Tracker* to detect.

An increase of the time resolution of the video (frames per second) would also reduce the other large source of uncertainty, which is in the period. For example, using a standard 60 fps camera, the uncertainty would be less than half of the current uncertainty, as not only will the time between frames halve, but also the uncertainty on exactly which frame contains a maximum displacement value will be lower, lowering the overall uncertainty.

## 5 CONCLUSION

This lab showed that the relationship between the period and release angle of a pendulum was quadratic, but could be approximated as constant with small angles. This did not match linear model that was provided in (Wilson, 2024). The initial Q factor was determined to be  $Q = 156$ , although there is significant disagreement between the methods provided in (Wilson, 2024). It was determined that the more consistent method was the line of best fit method. It was also found that the relationship between the period and length was exponential, although the numerical value of the exponent, 0.39, was in disagreement with the theoretical value of 0.5. There was no experimentally determined relationship between the Q factor and the length of the pendulum. To minimize error in future experiments, the best potential improvement to the set up include having a more rigid pivot point to reduce energy loss. To reduce uncertainty in data collection, using a better camera would help, as it allows for more precise position detection, as well as a better time resolution, reducing the largest uncertainty in the period.

## 6 BIBLIOGRAPHY

Brown, D. (2024). Tracker 6 Help. physlets.org.

Wilson, B. (2024). PHY180-PendulumProject. University of Toronto.

### 6.1 Appendix A

All the recorded data and the python code can be found at  
<https://github.com/jadontsai/phy180pendulumproject/tree/main>

### 6.2 Appendix B

AI assistance statement AI was used in this assignment to write a python script to make a better-looking amplitude-time graph. The AI used was Google's Gemini Advanced (1.5 pro) in the early morning hours of 04 October. The initial prompt was "Can you take these data points and plot them into a graph (dataset attached)". Helper prompts ranged from "can you please make it so that the t (x axis) starts counting at 0" and "can you instead do it where its just an independent counting function and not dependent on the line of best fit" and "the function is missing a few of the maxima points, notably near the beginning and at the end". The code is mostly AI generated, with light editing and personalization made to make it functional. No other AI assistance was used in writing the actual lab report